

Polygenic overlap when one trait is oligogenic

Eleven trait pairs across thirteen traits, one QC arm.
bipred 0.3.6, ldpred3 0.4.5, numpy 2.4.6, Windows-11 x86-64.
Estimates in `results/estimates.csv`.

The question, and the finding

Genome-wide genetic correlation averages over a million variants. For a trait whose heritability sits in one or two loci, that average is dominated by the $\sim 999,900$ variants doing nothing. The question here is whether bipred's polygenic-overlap parameterisation recovers a relationship that `rg` dilutes to noise.

Lipoprotein(a) is the case that makes it matter: essentially monogenic at *LPA*, a validated causal risk factor for coronary artery disease, and the target of phase-3 lowering trials. If `rg` cannot see that relationship, it is the wrong summary for a whole class of clinically important exposures.

It cannot. Fitted against CAD over 958,848 variants, Lp(a) has a genome-wide `rg` of $+0.0481$, and cross-trait LD Score regression puts the same quantity at -0.0945 (se 0.1161) — pointing the wrong way, and within one standard error of zero either way. The same fit estimates 106 causal variants for Lp(a) against 1,459 for CAD, 49 of them shared, with effects inside that shared component correlating $+0.3644$.

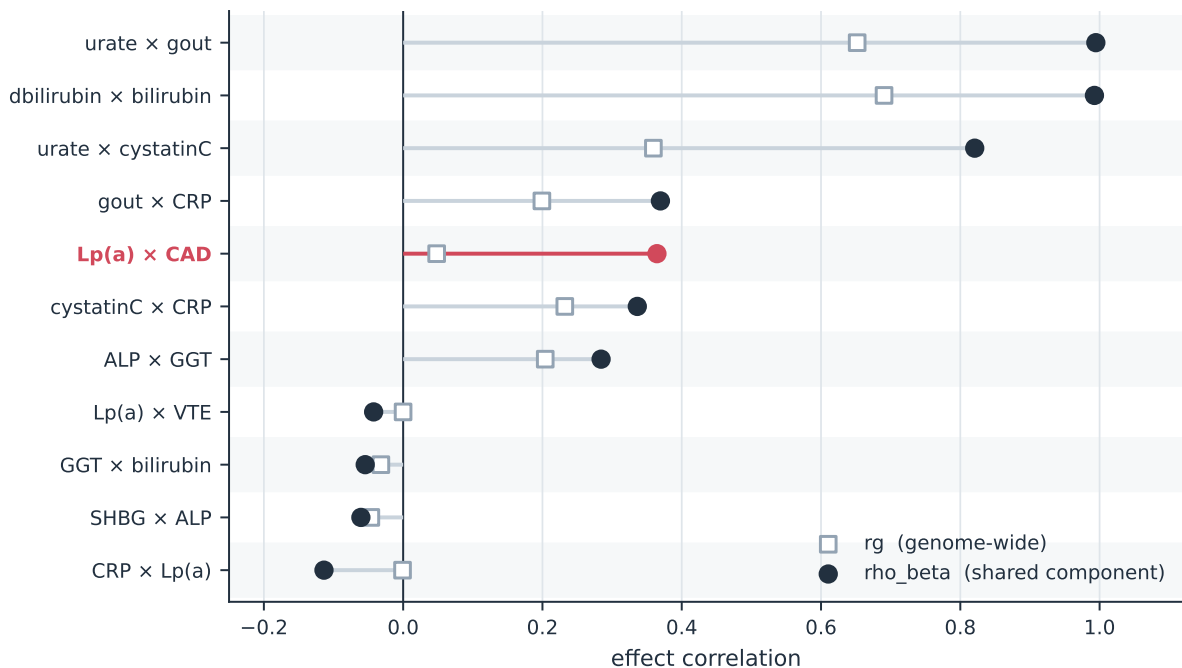


Figure 1: Pairs ordered by `rho_beta`, with genome-wide `rg` on the same row. The order is biological: same-biology and causal pairs at the top, unrelated pairs at the bottom. `rho_beta` exceeds `rg` almost everywhere, because `rg` is attenuated by the variants that carry no signal for either trait. What distinguishes Lp(a) $\times$ CAD is the size of the gap: its `rg` sits with the unrelated pairs while its `rho_beta` sits with the related ones. Compare gout $\times$ CRP directly above it — nearly the same `rho_beta` ($+0.3694$ against $+0.3644$) but four times the `rg` ($+0.1992$ against $+0.0481$).

The two pairs at the top are there because their answers were known in advance: direct against total bilirubin is two studies of the same biology, and urate against gout has crystal deposition as the disease mechanism. Both land near $+0.99$, which is what fixes the top of the scale. Four pairs sit below zero, between -0.11 and -0.04 — three chosen because no shared biology was expected ($\text{GGT} \times \text{bilirubin}$, $\text{SHBG} \times \text{ALP}$, $\text{CRP} \times \text{Lp(a)}$), plus $\text{Lp(a)} \times \text{VTE}$. Without both ends of that scale, $+0.3644$ would be a number with nothing to compare it to.

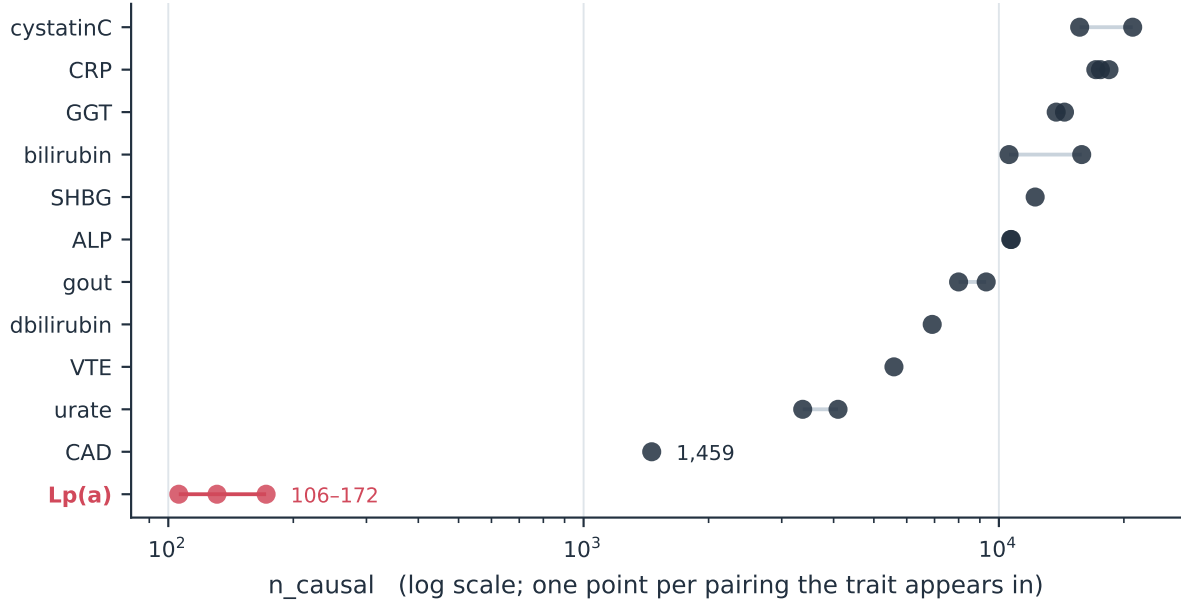


Figure 2: Estimated polygenicity per trait, one point per pairing the trait appears in. Lp(a) is estimated at 106, 131 and 172 causal variants across three independent pairings; every other trait here is in the thousands to tens of thousands. This is the arithmetic behind the finding — averaging one very large locus over $\sim 10^6$ variants destroys the signal, and the architecture summary does not average it away.

frac_shared is not the informative statistic

A second result fell out of the pairs included only to fix the null: sharing causal *variants* is nearly free, and carries little information on its own.

$\text{GGT} \times \text{bilirubin}$ and $\text{SHBG} \times \text{ALP}$ were chosen as unrelated pairs of polygenic traits. They share 63% and 71% of the smaller causal set — and their shared effects correlate at -0.05 and -0.06 . Sitting at nearly the same **frac_shared**, $\text{gout} \times \text{CRP}$ and $\text{cystatinC} \times \text{CRP}$ correlate at $+0.37$ and $+0.34$. For traits that are each causal at tens of thousands of variants, substantial overlap of the causal *sets* is close to the default, and it is the *direction* of the shared effects that discriminates.

The qualification matters: over the whole range **frac_shared** is not uninformative — it runs from 0.39 at the bottom of the scale to 1.00 at the top. What it cannot do is separate pairs in the middle, which is where most real pairs fall.

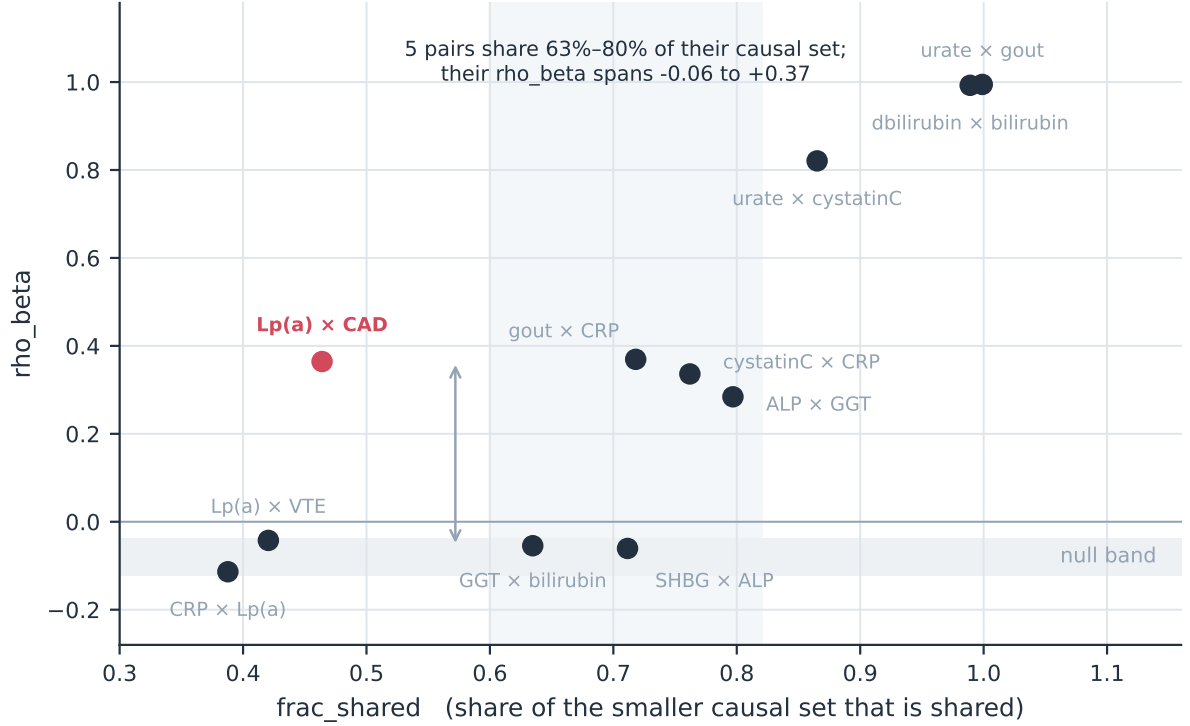


Figure 3: **frac_shared** against **rho_beta**. Across the full range the two are plainly associated — it is the middle that cannot be resolved. Inside the shaded column, five pairs share 63–80% of their causal set while their **rho_beta** spans -0.06 to $+0.37$. **Lp(a) × CAD** sits low on the horizontal axis because **Lp(a)** has few causal variants at all, which is the point: a small shared set with a consistent direction. The horizontal band marks where the four below-zero pairs fall; it is descriptive, not a test.

This matters for how the overlap summary should be read. **n_shared** and **frac_shared** describe how much of the architecture is common; **rho_beta** describes whether the common part agrees. Only the second distinguishes the pairs here.

Methods

The model

bipred is bivariate LDpred: one joint fit of two traits that share a single LD reference. Each variant is assigned to one of four latent states — causal for neither trait, for trait 1 only, for trait 2 only, or for *both* — with probabilities $(\pi_{00}, \pi_{10}, \pi_{01}, \pi_{11})$. A trait-1-only effect is drawn from $N(0, s_1)$, a trait-2-only effect from $N(0, s_2)$, and a both-causal pair from $N(0, \Sigma)$ with $\Sigma = \begin{bmatrix} s_1 & s_{12} \\ s_{12} & s_2 \end{bmatrix}$. The off-diagonal s_{12} is the effect covariance inside the shared component, and is the only place the two traits couple.

The indicator is *per trait*, not a single shared one. Whether the two traits’ causal variants co-occur is therefore learned as π_{11} rather than assumed: correlated traits borrow strength through the both-causal component, and disjoint traits can drive π_{11} toward zero so the two fits decouple. That is what makes the polygenic-overlap summary an estimate rather than a restatement of the prior.

Fitting is Gibbs sampling over the LD blocks. Each sweep evaluates the four bivariate-Gaussian likelihoods of the residual marginal estimate at a variant, samples that variant’s state, then draws its effects; the mixture weights π and the variance components (s_1, s_2, s_{12}) are re-estimated every sweep under a damping factor of 0.2. Reported quantities are posterior means over the retained iterations: per-trait SNP heritability **h2**, the genetic correlation **rg**, posterior-mean effects, and a MiXeR-style overlap summary — **n_causal** per trait, **n_shared**, **frac_shared**, and the correlation **rho_beta** of effects within the shared component.

What **rho_beta** is, and is not

rg is a genome-wide average: genetic covariance divided by the geometric mean of the two heritabilities, over every variant in the fit. **rho_beta** is the effect correlation *conditional on a variant being causal for both traits*. The two answer different questions, and they come apart exactly when one trait’s heritability sits in very few loci. Neither is a causal estimate.

frac_shared is π_{11} divided by the smaller of the two marginal causal fractions, so it asks what proportion of the less polygenic trait’s causal set is also causal for the other.

Sample overlap

Overlapping samples correlate the two traits’ sampling errors. The fit accepts that correlation as **cross_corr**, set per pair from the cross-trait LD Score regression intercept, used as-is and never clipped into range. An intercept can also reflect correlated confounding rather than shared samples, so this absorbs overlap without identifying it.

The LD-consistency screen

Per-variant filters — minor allele frequency, imputation quality, a chi-square cap, sample size — judge each variant in isolation, so none of them can see a variant disagreeing with the variants it is correlated with. The screen tests exactly that. Within a window the variants are split at random in half and each z -score in one half is predicted from the other through the LD,

$$\hat{z}_a = R_{aB} R_{BB}^+ z_B, \quad T_a = \frac{(z_a - \hat{z}_a)^2}{1 - R_{aB} R_{BB}^+ R_{Ba}} \sim \chi_1^2$$

under consistency. Variants far from what their neighbours predict are dropped, and the split is repeated so each variant is tested from several directions.

It runs against the same numeric LD representation the fit will use, so the screen and the sampler see the same matrix. It is DENTIST-inspired, not DENTIST: window construction, partition schedule, eigenvalue regularisation and removal policy all differ from the published pipeline (Chen et al. 2021, *Nature Communications* 12:7117), so that pipeline’s calibration does not transfer.

Divergence diagnostics

A bivariate fit can diverge silently. Given summary statistics that disagree with the LD reference, the sampler places large opposing effects on variants in near-perfect LD, and these cancel to a plausible-looking **h2**. Three diagnostics are recorded, and any one of them raises the warning: $\sum \beta^2/h^2$ above 10 (the cancellation ratio, reported in both studies), $\max |\beta|$ above 25 slab

standard deviations, and retained-trace drift above 1.25. All three were calibrated on one real LDL $\times$ CAD fit that diverged and the same fit after screening repaired it, so each separates the two regimes by more than an order of magnitude rather than by a hair.

An estimate carrying the divergence warning is not an estimate of anything, and is reported only to show the contrast.

LD reference and harmonisation

All fits use the European UK Biobank HapMap3 LD reference, built with ldpred3's bigsnpr converter at the pinned revision 5d86ac9d. Summary statistics are harmonised to it by rsID with allele matching, flipping the effect sign where the effect and reference alleles are swapped; strand-ambiguous and unmatched variants are dropped. Odds-ratio releases are taken to logs before harmonisation, which preserves the sign across a flip. Case/control traits are fitted at an effective N of $4/(1/n_{\text{case}} + 1/n_{\text{ctrl}})$. BLAS is pinned to one thread throughout.

This analysis

Every pair was fitted under a single QC arm, **001**: lenient per-variant filters ($\text{MAF} \geq 0.01$, imputation $\text{INFO} \geq 0.9$, $\chi^2 \leq 80$, allele-frequency discordance against the reference ≤ 0.20), long-range LD regions and the MHC *retained*, and the LD-consistency screen *on*, four rounds. That arm was chosen because the companion **ldl-cad** study found the screen to be the load-bearing switch of the three; fixing the other two isolates it.

The screen is run once per trait on the lenient variant set, and the resulting masks are intersected across the pair — not re-run per pair. Re-screening per pair would let the screen see a different variant set each time, which entangles what the per-variant filters did with what the screen was given.

Traits, source files, sample sizes, ancestries and known dataset defects are registered in `../_lib/traits.toml`. Sources that were tried and rejected are recorded with their symptoms in `../_lib/datasets.md`; read that before adding a dataset, because two of the three defects recorded there are invisible until a fit has already started.

This is one QC arm on one LD reference. It measures how the two summaries behave across these pairs; it is not a comparison of QC strategies, and the factorial that would support one is in `benchmarks/qc_factorial.py`.

Full results

The figures above are a reading of one table, not a substitute for it.

<code>results/estimates.csv</code>	one row per pair: <code>m</code> , LDSC and bipred <code>rg</code> , <code>h2</code> , the overlap parameters, cancellation ratios, divergence flag
<code>results/provenance.json</code>	package versions, platform, clean source revisions
<code>../_lib/traits.toml</code>	every trait: file, column layout, N , ancestry, known quirks
<code>../_lib/datasets.md</code>	sources tried and rejected, with symptoms

Thirteen rows are recorded: the eleven fitted pairs above, and two fits of bilirubin $\times$ gallstones that were rejected on data grounds rather than on their results. The gallstones release carries

minor allele frequency under an effect-allele-frequency header, and the refit diverged (cancellation 14.29). Both rows are kept in the CSV so the rejection is auditable; `../_lib/datasets.md` records how it was diagnosed.

Summary statistics are not committed — roughly 9 GB across the thirteen traits. Nothing in this report requires them: `figures.py` reads only `results/estimates.csv`.

Caveats

- **The overlap statistics carry no interval.** The null band is descriptive, not a test, so “+0.3644 is outside it” is an observation about eleven fits and not an inference. This is the largest gap in the analysis.
- **$rg = +0.0481$ for $Lp(a) \times CAD$ is not a null result**, and will be read as one if it is quoted without the architecture that explains it.
- **Every oligogenic reading here is $Lp(a)$.** Its `n_causal` reproduces across three pairings (106, 131, 172), which is reassuring about the estimate but says nothing about whether the effect generalises to other oligogenic traits.
- **$Lp(a)$ ’s `h2` of 0.007 is a reference limitation.** HapMap3 tags the *LPA* KIV-2 repeat poorly. It is not an estimate of $Lp(a)$ heritability, and the shared-component correlation is measured on what the reference can see.
- **$Lp(a)$ is a four-cohort meta-analysis at 95.9% European** fitted against a European LD reference. The European-only alternative covered 16% of the reference and was unusable.
- **Sample overlap is substantial** — most exposures are UK Biobank. The cross-trait LDSC intercept absorbs it and is recorded per pair, but an intercept cannot identify overlap on its own.
- **The Sinnott-Armstrong panel publishes no allele frequency**, so the AF-concordance check is vacuous for those traits and the +1.0000 it reports is an artifact of comparing a column with itself.